Friday 16 Oct: presenter's card

slides: `meeting_plots/oct16/slides.pdf`

Open with: "T1 is complete in 17 regions. I built the T3 test harness and finished T2 tier 1. The headline: terrain from elevation data predicts hiking speed in regions the model has never seen; OSM trail tags barely do. It survived every check I could think of."

Numbers to know

- **Data:** 17 regions, 2,876 hiking traces; walking set 1,992 outings, 290,640 segments of 50 m, 4,702 h.
- **Tobler:** 13% worse than our curve. Real hikers peak near **4.2 km/h**, not 6.
- **Unseen region** (skill vs slope curve): terrain **+2.1%** (trees +2.8%), tags +0.9% (+0.4%). Terrain beats tags in **16 of 17** regions.
- **Harder tests:** a whole zone left out +1.9%; West ↔ East +2.6%; newer hikes +2.2%; same at every GPS rate.
- **Whole hike:** time error 13.6% → 9.6% with the hiker's own pace (first km) + terrain.
- **Same hiker, same trail:** down is 8% faster than up on gentle grades, 22% on steep ones. Tobler says 42%.
- **Modes:** runners = walkers ×2.5; bikes need their own curve.

New since last week

- Why tags fail: not different meanings (within a hike, harder sac_scale is slower in every region). They're missing on much of the trail (1–96% by region).
- Sparse GPS traces read up to 11% slow: how we measure, not slower hikers.
- A 2026 paper (Huang et al., Hong Kong) uses boosted trees with weather, one city, random split: no transfer test.
- The 5-number formula travels less well than the full model (+0.6% for an unseen zone). Say so.

Ask Danni

1. Framing: "terrain transfers, tags don't". Agree?
2. Embeddings: keep as "helps locally, transfers less"?
3. Lidar understory: ∼12 GB for the White Mountains. Do it?
4. Data release: license (ODbL) and lab policy.
5. Cluster account: `engr-class-any` or the lab's?
6. Fix sparse traces in a T1 v2, or just report it?
7. Which 3–4 figures to keep (the draft is at 5 pages).

Same order as slide 14.

Hard questions, short answers

"Isn't +2% small?"

Per 50 m, yes: GPS noise and the hiker dominate. But it's consistent (17 of 17 regions in direction), its interval excludes zero, and it's more than twice what tags give.

"Is terrain just a stand-in for the region?"

No: it adds ∼1.9 points over each region's own average, and holds with a whole zone left out.

"Why no deep learning or images?"

About 2,000 outings is small. We tested frozen satellite embeddings: +1.1% in new regions, less reliable than terrain.

"What about weather?"

Tested (Daymet, each hike's day and place): hot days are 3% slower and snow 4%, but weather doesn't improve predictions in new regions; the hiker's own pace already holds it. Same for date and time of day.

"What did the AI do?"

Claude helped write and run the code and draft the documents; I review the work and I'm responsible for it.